

## BASIC SQL QUESTIONS (1–15)

### 1. What is SQL?

**Answer:**

SQL (Structured Query Language) is used to store, retrieve, update, and manage data in relational databases.

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### 2. What are types of SQL commands?

**Answer:**

- **DDL** – CREATE, DROP, ALTER
  - **DML** – INSERT, UPDATE, DELETE
  - **DQL** – SELECT
  - **DCL** – GRANT, REVOKE
  - **TCL** – COMMIT, ROLLBACK
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### 3. Difference between WHERE and HAVING?

**Answer:**

- WHERE filters rows before grouping.
  - HAVING filters after GROUP BY.
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### 4. Difference between DELETE, TRUNCATE, DROP?

| DELETE                | TRUNCATE                 | DROP                   |
|-----------------------|--------------------------|------------------------|
| Removes specific rows | Removes all rows         | Deletes entire table   |
| Can use WHERE         | No WHERE                 | Removes structure also |
| Can rollback          | Cannot rollback (mostly) | Cannot rollback        |

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### 5. What is a Primary Key?

**Answer:**

A column that uniquely identifies each row. It cannot contain NULL values.

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## 6. What is a Foreign Key?

### Answer:

A column that links one table to another table's primary key.

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## 7. Difference between CHAR and VARCHAR?

### Answer:

- CHAR → Fixed length
  - VARCHAR → Variable length
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## 8. What is NULL?

### Answer:

A NULL represents missing or unknown value.

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## 9. What is DISTINCT?

### Answer:

Removes duplicate values.

SELECT DISTINCT city FROM customers;

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## 10. GROUP BY vs ORDER BY?

- GROUP BY → Groups rows
  - ORDER BY → Sorts rows
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## 11. What are Constraints?

### Answer:

Rules applied to columns like:

NOT NULL, UNIQUE, PRIMARY KEY, FOREIGN KEY, CHECK.

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## 12. What is AUTO\_INCREMENT?

**Answer:**

Automatically generates unique numbers.

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**13. What is Composite Key?**

**Answer:**

Primary key made of two or more columns.

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**14. Default order of ORDER BY?**

**Answer:**

Ascending (ASC).

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**15. How to rename a table?**

RENAME TABLE old\_name TO new\_name;

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◆ **INTERMEDIATE QUESTIONS (16–30)**

**16. What is JOIN?**

**Answer:**

Combines rows from multiple tables.

Types:

- INNER JOIN
  - LEFT JOIN
  - RIGHT JOIN
  - FULL JOIN
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**17. INNER JOIN vs LEFT JOIN?**

**Answer:**

- INNER → Only matching rows
  - LEFT → All left table rows + matched right rows
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## **18. What is Self Join?**

### **Answer:**

Joining a table with itself.

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## **19. What is Subquery?**

### **Answer:**

Query inside another query.

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## **20. Subquery vs Join?**

### **Answer:**

Join is usually faster and better for combining tables. Subquery is nested inside another query.

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## **21. What is a View?**

### **Answer:**

Virtual table created from SELECT query.

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## **22. What is Index?**

### **Answer:**

Improves query performance by faster searching.

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## **23. What is Normalization?**

### **Answer:**

Process of reducing data redundancy.

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## **24. Explain 1NF, 2NF, 3NF?**

- **1NF:** No repeating groups
  - **2NF:** No partial dependency
  - **3NF:** No transitive dependency
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## **25. What is Denormalization?**

### **Answer:**

Combining tables to improve performance.

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## **26. What is Stored Procedure?**

### **Answer:**

Pre-written SQL code stored in database.

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## **27. What is Trigger?**

### **Answer:**

Automatically executes when event occurs.

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## **28. What is Cursor?**

### **Answer:**

Used to process rows one by one.

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## **29. What is Transaction?**

### **Answer:**

Group of SQL statements executed together.

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## **30. What is ACID?**

- **A** – Atomicity
  - **C** – Consistency
  - **I** – Isolation
  - **D** – Durability
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## **◆ ADVANCED QUESTIONS (31–40)**

## **31. UNION vs UNION ALL?**

- **UNION** → Removes duplicates

- UNION ALL → Keeps duplicates
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### **32. What is Correlated Subquery?**

**Answer:**

Subquery that depends on outer query.

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### **33. What is CTE?**

**Answer:**

Temporary result set using WITH.

```
WITH temp AS (  
SELECT * FROM employees  
)  
SELECT * FROM temp;
```

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### **34. RANK() vs DENSE\_RANK()?**

- RANK → Skips rank numbers
  - DENSE\_RANK → No skipping
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### **35. What are Window Functions?**

**Answer:**

Functions that perform calculations across rows (RANK, SUM OVER).

### **36. Find Second Highest Salary?**

```
SELECT MAX(salary)  
FROM employees  
WHERE salary < (SELECT MAX(salary) FROM employees);
```

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### **37. Remove duplicate rows?**

```
SELECT DISTINCT * FROM table_name;
```

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### **38. Types of Index?**

- Clustered
  - Non-clustered
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### **39. What is Query Optimization?**

**Answer:**

Improving performance of SQL queries.

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### **40. What is Execution Plan?**

**Answer:**

Shows how SQL query is executed internally.

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### **◆ PRACTICAL QUESTIONS (41–50)**

#### **41. Top 5 highest salaries?**

```
SELECT * FROM employees  
ORDER BY salary DESC  
LIMIT 5;
```

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#### **42. Salary > Average salary?**

```
SELECT * FROM employees  
WHERE salary > (SELECT AVG(salary) FROM employees);
```

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#### **43. Find duplicate records?**

```
SELECT name, COUNT(*)  
FROM employees  
GROUP BY name  
HAVING COUNT(*) > 1;
```

---

#### **44. Delete duplicate records?**

```
DELETE e1 FROM employees e1  
JOIN employees e2
```

ON e1.id > e2.id  
AND e1.name = e2.name;

---

**45. Nth highest salary?**

SELECT DISTINCT salary  
FROM employees  
ORDER BY salary DESC  
LIMIT 1 OFFSET 2;

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**46. Employees joined last 30 days?**

SELECT \*  
FROM employees  
WHERE join\_date >= CURDATE() - INTERVAL 30 DAY;

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**47. Count employees in each department?**

SELECT department, COUNT(\*)  
FROM employees  
GROUP BY department;

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**48. Departments with more than 5 employees?**

SELECT department  
FROM employees  
GROUP BY department  
HAVING COUNT(\*) > 5;

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**49. Employees without manager?**

SELECT \*  
FROM employees  
WHERE manager\_id IS NULL;

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### **50. Query using 3 tables?**

```
SELECT e.name, d.department_name, p.project_name  
FROM employees e  
JOIN departments d ON e.department_id = d.id  
JOIN projects p ON e.project_id = p.id;
```